YOLO V8
安装与配置教程

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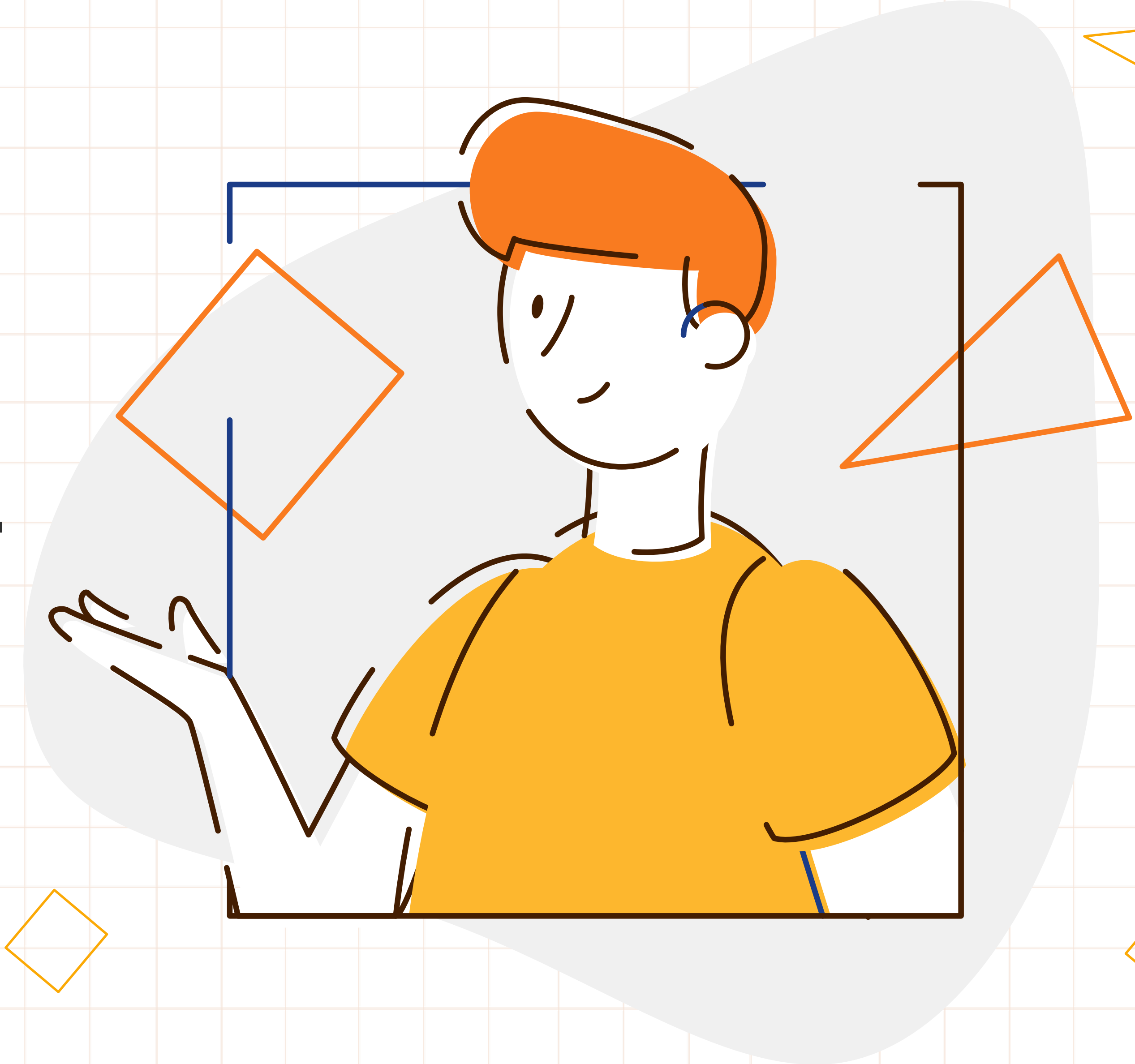

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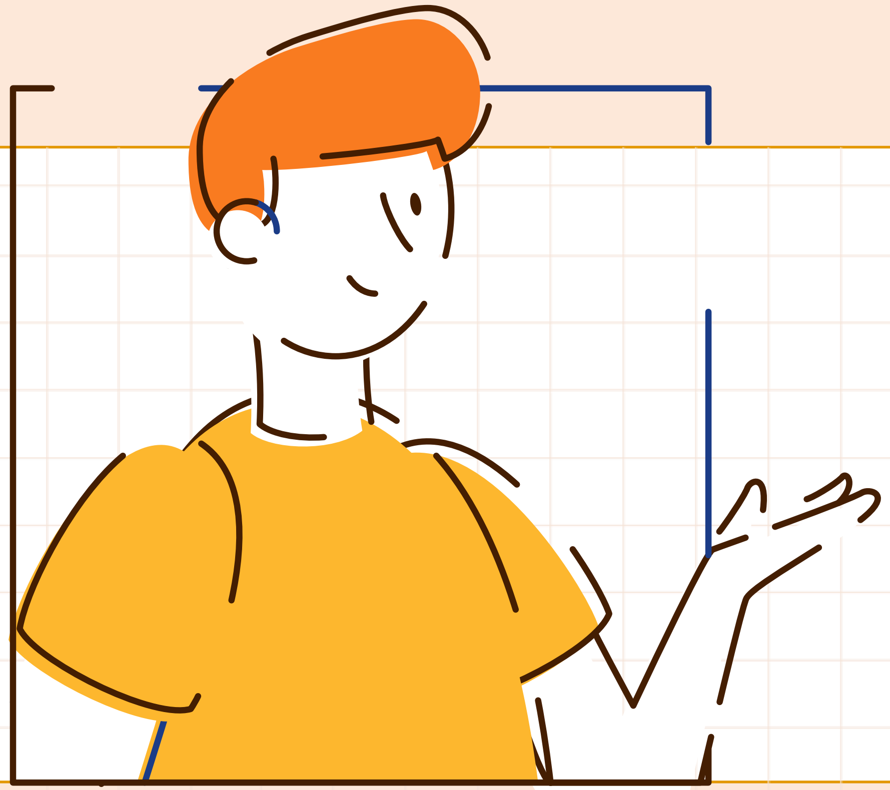


STEP 01

创建虚拟环境

创建虚拟环境

- 如果不是专用的开发环境推荐使用conda创建虚拟环境

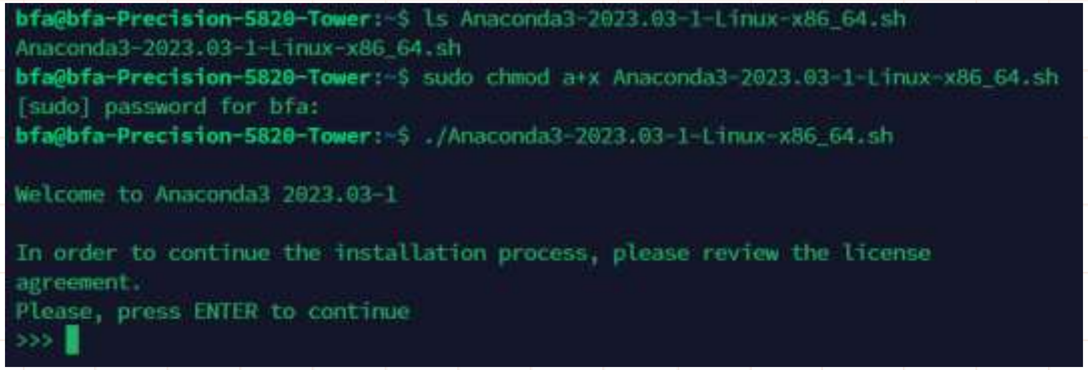
linux下安装conda

- 查找需要的版本：https://repo.anaconda.com/archive/
- 下载，这里以x86的机器为例：

```
bfa@bfa-Precision-5820-Tower:~$ wget https://repo.anaconda.com/archive/Anaconda3-2023.03-1-Linux-x86_64.sh
--2023-06-18 22:01:57--  https://repo.anaconda.com/archive/Anaconda3-2023.03-1-Linux-x86_64.sh
Resolving repo.anaconda.com (repo.anaconda.com)... 2606:4700::6810:8303, 2606:4700::6810:8203, 104.16.131.3, ...
Connecting to repo.anaconda.com (repo.anaconda.com)|2606:4700::6810:8303|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 902411137 (861M) [application/x-sh]
Saving to: 'Anaconda3-2023.03-1-Linux-x86_64.sh'

Anaconda3-2023.03-1-Linux-x86_64.sh       100%[=================================================================================>] 860.61M  74.7MB/s    in 11s

2023-06-18 22:02:08 (75.3 MB/s) - 'Anaconda3-2023.03-1-Linux-x86_64.sh' saved [902411137/902411137]

bfa@bfa-Precision-5820-Tower:~$
```

创建虚拟环境

- 更改文件执行权限
- 以用户身份执行.sh文件
- 回车或yes直到安装完毕

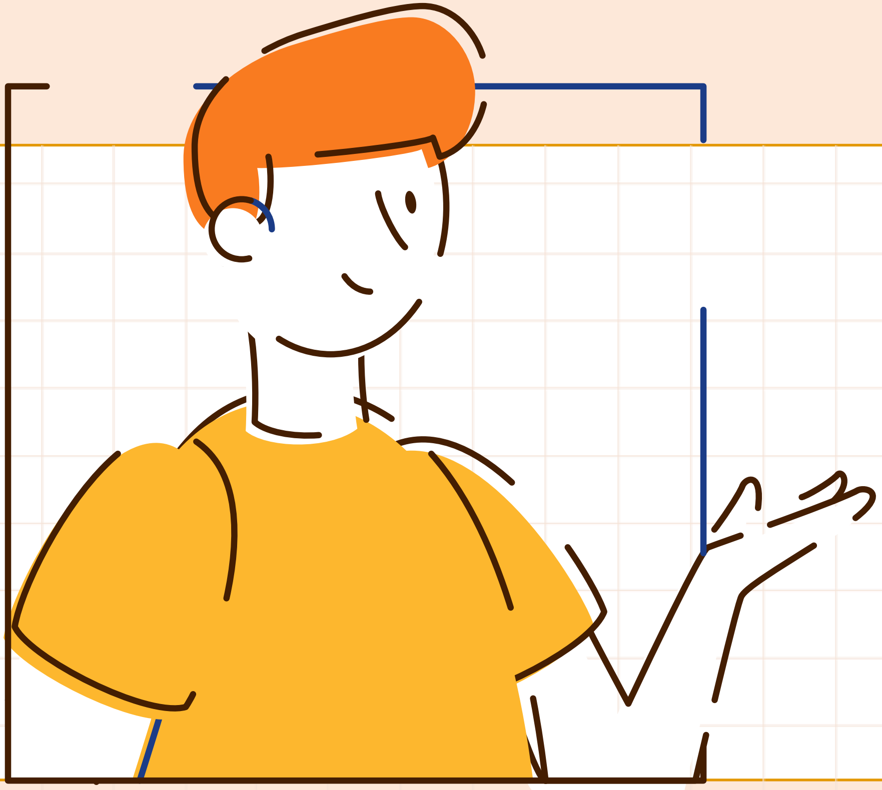


创建虚拟环境

- 重启终端，看到出现"（base）"代表安装成功
- 使用conda创建新的虚拟环境，这里命名为yoloV8
- 使用conda env list 查看所有虚拟环境

```
New release '22.04.2 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Your Hardware Enablement Stack (HWE) is supported until April 2025.
Last login: Sun Jun 18 22:01:01 2023 from 10.128.138.214
(base) bfa@bfa-Precision-5820-Tower:~$ conda create -n yoloV8 python=3.10
Collecting package metadata (current_repodata.json): done
Solving environment: done


==> WARNING: A newer version of conda exists. <==
  current version: 23.3.1
  latest version: 23.5.0

(base) bfa@bfa-Precision-5820-Tower:~$ conda env list
# conda environments:
#
base                  *  /home/bfa/anaconda3
yoloV8                   /home/bfa/anaconda3/envs/yoloV8

(base) bfa@bfa-Precision-5820-Tower:~$ 
```

- Windows 安装conda可以直接去anaconda官网下载安装
- 建议安装时不要勾选将conda加入环境变量，安装完成后如有需要可以手动加到环境变量。



STEP 02

环境配置及演示

环境配置

- 激活虚拟环境
- 安装pytorch GPU版本：https://pytorch.org/get-started/locally/
- 使用pip安装ultralytics

```
(base) bfa@bfa-Precision-5820-Tower:~$ conda activate yoloV8
(yoloV8) bfa@bfa-Precision-5820-Tower:~$ pip install ultralytics
```

torch_check.py

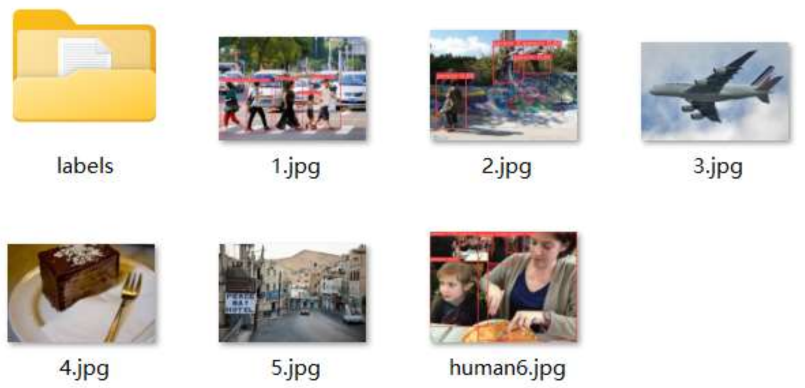
```python
import torch
print(torch.__version__)
print(torch.cuda.is_available())
```

- 以VSCode为平台演示yoloV8的用法，建议在宿主机器上装一下VPN，这样下载github上的东西方便一些。
- Demo演示
- 官网文档链接：https://docs.ultralytics.com/
- Github链接：https://github.com/ultralytics/ultralytics

demo演示

- 推理单张图片或多张图片
- Img_path既可以是图片目录，也可以是具体的某一张图片路径
- Conf表示置信度，即模型有多大的把握预测结果是对的，取值0-1
- Classes表示希望检测的类别编号，具体有哪些类别既可以参考dataset.yaml，也可以print(result)查看，classes=[0,2,3]表示同时检测编号0、2、3代表的类。
- Save_txt=True可以把检测结果以txt保存下来，其中txt的文件名与图片的文件名一一对应，txt中每行代表一个检测框，从左至右的数字以此代表类别号、边界框中心点x、y坐标和边界框宽度和高度。

```
img_path = "./image/"
results = model.predict(img_path, save=True,conf=0.5) # device=0 by default, conf:置信度阈值
results = model.predict(img_path,save=True,classes=0,conf=0.4) # i.e. classes=0,classes=[0,3,4]

# save detection results *
results = model.predict(img_path,save=True,save_txt=True,classes=0,conf=0.4)
```



名称	修改日期	类型
1.txt	2023/6/25 19:22	文本文档
2.txt	2023/6/25 19:22	文本文档
human6.txt	2023/6/25 19:22	文本文档

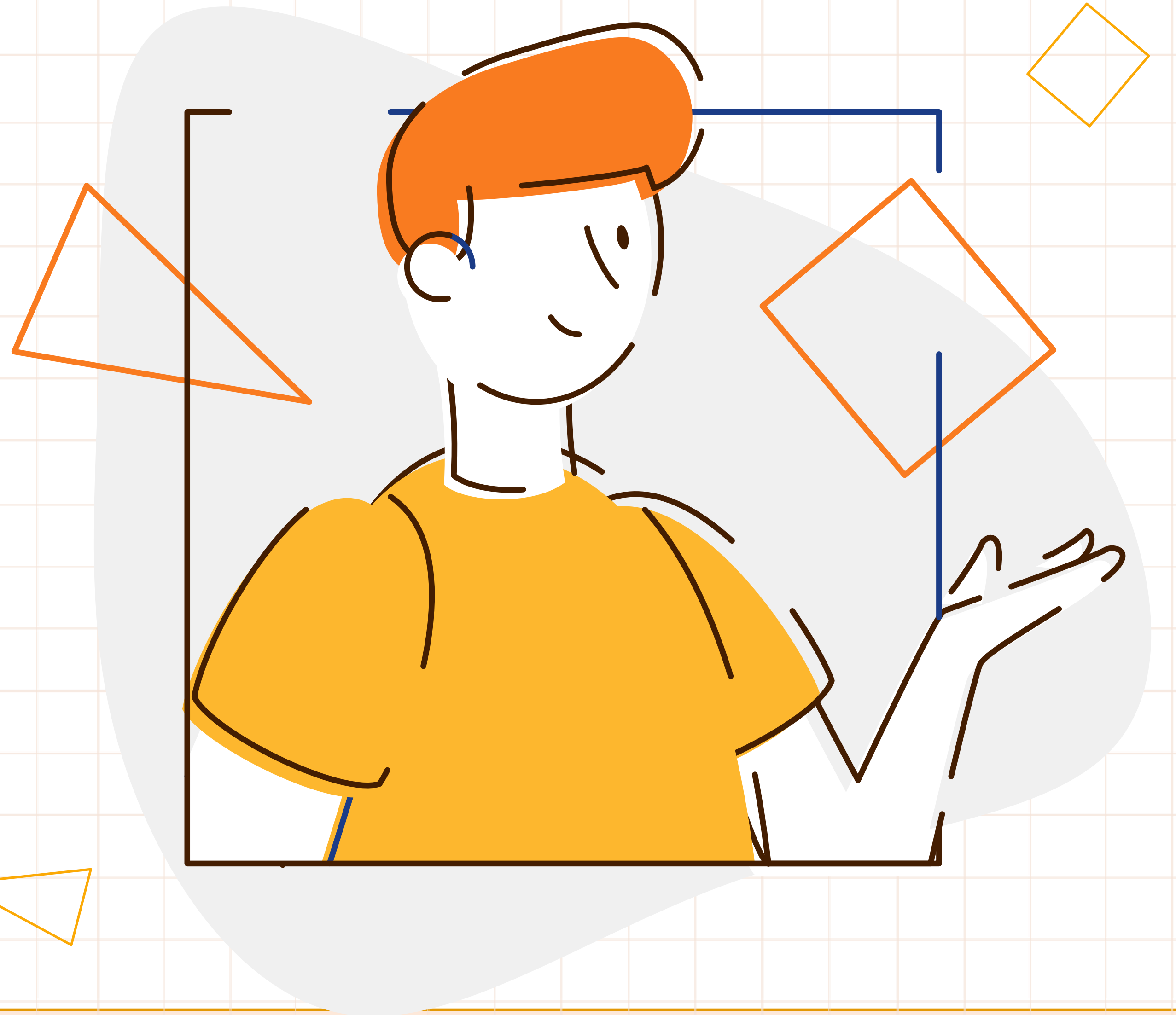
```
0 0.102443 0.687439 0.18112 0.508388
0 0.277851 0.677364 0.173231 0.477222
0 0.732145 0.653912 0.197108 0.495006
0 0.478738 0.662437 0.187275 0.484688
0 0.615477 0.730766 0.150116 0.345944
```

demo演示

- 推理视频或者usb摄像头
- Video_path表示视频路径
- 要"import cv2"才能使用cv2包
- 如果要通过摄像头传输视频进行推理，将cv2.VideoCapture(…)中的参数改为0（值为0而不是0字符）
- 在循环内同样可以对每帧进行处理，如对每帧进行resize，将结果逐帧另存为一个视频等等，具体操作可以查看cv2指南。

```
# predict video
video_path = "./video/1.mp4"
cap = cv2.VideoCapture(video_path)

# Loop through the video frames
while cap.isOpened():
    # Read a frame from the video
    success, frame = cap.read()

    if success:
        # Run YOLOv8 inference on the frame
        results = model(frame)

        # Visualize the results on the frame
        annotated_frame = results[0].plot()

        # Display the annotated frame
        cv2.imshow("YOLOv8 Inference", annotated_frame)

        # Break the loop if 'q' is pressed
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break
    else:
        # Break the loop if the end of the video is reached
        break

# Release the video capture object and close the display window
cap.release()
cv2.destroyAllWindows()
```

LOGO HERE



THANK YOU

感谢聆听

汇报人：OfficePLUS